

Phase 0 — Source System Analysis

AdventureWorks End-to-End BI Project — OLTP Database Analysis & Initial Star Schema Design

1. Goal & Methodology

The goal of this phase is to survey the AdventureWorks2025 sample OLTP database and identify which tables are suitable candidates for building a Data Warehouse with a Star Schema architecture. This analysis covers row-count profiling of candidate tables (to distinguish high-volume transactional tables from low-volume descriptive ones), a review of Foreign Key relationships between tables (to determine the correct join path), and an explicit list of data deliberately left out of this project's scope.

Row counts for candidate tables were obtained via SQL Server's own metadata (`sys.tables` and `sys.partitions`) rather than `COUNT(*)` on each table individually — faster and equally accurate.

Rule of thumb: Fact vs. Dimension

A table that records "something that happened" and holds additive numeric columns (price, quantity) is usually a high-volume Fact candidate. A table that holds descriptive, slower-changing attributes (product, customer, territory) is a Dimension candidate.

2. Schemas Reviewed

Schema	Key Tables
Sales	SalesOrderHeader, SalesOrderDetail, Customer, Store, SalesTerritory, SpecialOffer, Currency
Production	Product, ProductSubcategory, ProductCategory
Person	Person, Address, StateProvince, CountryRegion
Purchasing	PurchaseOrderHeader, Vendor
HumanResources	Employee, Department

3. Fact Table Candidates (Confirmed)

OLTP Table	Row Count	Conclusion
Sales.SalesOrderDetail	121,317	Primary Fact — one row = one order line with price and quantity
Sales.SalesOrderHeader	31,465	Supplemental info (date, customer, territory) to enrich the fact

4. Dimension Table Candidates (Confirmed)

Dimension	Source OLTP Table(s)
DimCustomer	Sales.Customer + Person.Person + Person.Address + Person.EmailAddress
DimProduct	Production.Product + ProductSubcategory + ProductCategory
DimReseller	Sales.Store
DimSalesTerritory	Sales.SalesTerritory
DimSalesPerson	Sales.SalesPerson + HumanResources.Employee + Person.Person
DimPromotion	Sales.SpecialOffer
DimCurrency	Sales.Currency
DimGeography	Person.Address + StateProvince + CountryRegion
DimDate	Manually generated (Date Generator Script) — no OLTP source

Note: DimCustomer is built from four separate OLTP tables (customer + personal details + address + email). This complexity is exactly why, later in Phase 3, its incremental extraction query had to be moved into a Stored Procedure (due to an SSIS OLE DB parameter-parsing limitation with CTEs).

5. Data Deliberately Left Out of Scope

The following tables exist in the OLTP database but were intentionally excluded from this project's scope (Internet/Reseller sales analysis), either because they belong to a different subject area or would warrant an entirely separate fact table in a future extension:

- **ShoppingCartItem** — relates to in-progress cart activity, not historical sales analysis
- **CreditCard, PersonCreditCard** — payment details, outside the scope of sales analysis
- **TransactionHistory, WorkOrder, WorkOrderRouting, BillOfMaterials** — related to manufacturing/inventory; could become a future FactProduction in a later extension of this project
- **Document, Illustration, ProductPhoto, ProductReview** — unstructured content (files/text), outside the dimensional model

6. Foreign Key Review (sys.foreign_keys)

The main join path for building the fact tables was derived from the following FK chain:

```
SalesOrderDetail -> SalesOrderHeader -> Customer -> Person / Store
-> SalesTerritory
-> SalesPerson (Reseller only)
SalesOrderDetail -> SpecialOfferProduct -> SpecialOffer, Product
```

Internet Sales vs. Reseller Sales split rule

Whether an order belongs to the Internet channel (direct/individual customer) or the Reseller channel (store/B2B) is determined by the value of `Customer.StoreID` (linked to Store via `FK_Customer_Store_StoreID`):

Condition	Channel
Customer.StoreID IS NULL	Internet Sales — direct/individual customer
Customer.StoreID IS NOT NULL	Reseller Sales — store/B2B

- `SpecialOfferProduct` is the bridge table between `Product` and `SalesOrderDetail`, providing the `SpecialOfferID` used to isolate discounts.
- The `Person` → `Employee` → `SalesPerson` chain is only needed to resolve `DimSalesPerson` and only the Reseller side connects to it.
- The `SalesTerritory` → `BusinessEntityAddress` → `Address` → `StateProvince` → `CountryRegion` chain is the basis for the `DimGeography` design and the geography columns of `DimCustomer`.

7. Initial Star Schema Design

Based on the analysis above, the initial Data Warehouse structure was set as follows:

Fact Tables

FactInternetSales
FactResellerSales

Dimension Tables

DimDate, DimProduct, DimCustomer, DimReseller, DimSalesTerritory, DimSalesPerson,
DimPromotion, DimCurrency, DimGeography

Conformed Dimensions — sharing matrix between the two facts

This matrix identifies which dimension connects to which fact table — the basis for a later architecture decision to use a single, unified Data Warehouse (Kimball Bus Architecture) instead of separate Data Marts (see ARCHITECTURE.md).

Dimension	FactInternetSales	FactResellerSales
DimDate	Yes	Yes
DimProduct	Yes	Yes
DimSalesTerritory	Yes	Yes
DimPromotion	Yes	Yes
DimCurrency	Yes	Yes
DimCustomer	Yes	—
DimReseller	—	Yes
DimSalesPerson	—	Yes

DimGeography was planned in the initial design, but in practice (Phase 2 ETL) the geography columns ended up denormalized directly inside DimCustomer and DimReseller. This table was therefore excluded from the SSAS Tabular model in Phase 4 (full details in ARCHITECTURE.md, Decision #10).

Output of This Phase

Based on this analysis, the initial Data Warehouse DDL script (01_create_dw_schema.sql) was written in Phase 1 and became the foundation for every subsequent phase of the project (ETL, incremental load, the SSAS Tabular model, and the Power BI dashboard).